

Diseño Digital Avanzado

Instalación de Vitis 2024.2

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Instalación de Vitis/Vivado



Instalación de Vitis/Vivado

Xilinx - Vitis Core Development Kit - 2024.2

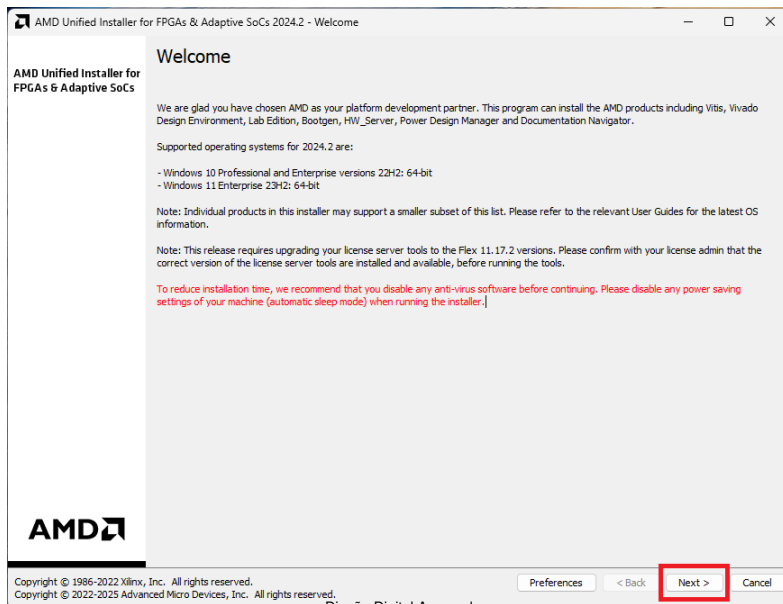
- Acceder al siguiente link y descargar el instalador web dependiendo del sistema operativo (Windows o Linux).
- <https://www.xilinx.com/support/download.html>
- Se debe crear un usuario en Xilinx para poder descargar el instalador.
(<https://www.xilinx.com/registration/create-account.html>)
- Después de descargar el instalador y ejecutarlo dependiendo del SO:
 - Windows: se puede ejecutar directamente el archivo
FPGAs_AdaptiveSoCs_Unified_2024.2_1113_1001_Win64.exe
 - Linux: se debe dar permisos de ejecución antes de ejecutar el instalador:
chmod +x FPGAs_AdaptiveSoCs_Unified_2024.2_1113_1001_Lin64.bin
sudo ./FPGAs_AdaptiveSoCs_Unified_2024.2_1113_1001_Lin64.bin
- Ver pasos de instalación en Pag. 5-12

Instalación de Vitis/Vivado

The screenshot shows the AMD website's 'Downloads' page. The navigation bar includes 'AMD', 'Products', 'Solutions', 'Resources & Support', and 'Shop'. The 'Downloads' section is active, showing a list of download categories: 'Vivado (HW Developer)', 'Vitis (SW Developer)', 'Vitis Embedded Platform', 'PetaLinux', 'Power Design Manager', 'Device Models', and 'Documentation Navigator'. The 'Vivado (HW Developer)' category is highlighted. Below this, the 'Version' section lists '2024.2' as the current version, with links to 'Vivado Archive', 'ISE Archive', and 'CAE Vendor Libraries Archive'. The main content area is titled 'Vivado™ Edition - 2024.2 Full Product Installation'. It contains 'Important Information' stating that Vivado™ 2024.2 is now available for download, 'Advanced Flow for Place-and-Route of All Versal™ Devices' with a list of benefits (Automatic partition-based placement and parallel P&R, Reduces congestion and improves routability for fast design closure, Default flow for all Versal devices), and 'Enabling Top-Level RTL Flows for Versal Devices' with a list of benefits (Configure key components like NoC and transceivers from top-level RTL, Enables programmable logic developers to stay in a RTL-centric design environment). A 'Download Includes' table lists the components: Vivado Design Suite (All Editions), Full Product Installation, Last Updated (Nov 18, 2024), Answers (2024.x - Vivado Known Issues), Documentation (Release Notes, OS Support Update, What's New in Vivado), and Support Forums (Installation and Licensing). A 'Feedback' button is located at the bottom right of the main content area.

The screenshot shows the AMD download verification page for the Vivado 2024.2 installer. It features a red box highlighting the download link: 'AMD Unified Installer for FPGAs & Adaptive SoCs 2024.2: Windows Self Extracting Web Installer (EXE - 222.91 MB)'. Below this, the MD5 SUM Value is provided: '84ee3816d8cbd0e0e38a1f9ac7eaa780'. The 'Download Verification' section includes a link to the installer and three buttons: 'Digests', 'Signature', and 'Public Key'. A second red box highlights the download link for the Linux version: 'AMD Unified Installer for FPGAs & Adaptive SoCs 2024.2: Linux Self Extracting Web Installer (BIN - 303.93 MB)'. Below this, the MD5 SUM Value is provided: '20c806793b3ea8d79273d5138bd195f'. The 'Download Verification' section also includes a link to the installer and three buttons: 'Digests', 'Signature', and 'Public Key'.

Instalación de Vitis/Vivado



Instalación de Vitis/Vivado

AMD Unified Installer for FPGAs & Adaptive SoCs 2024.2 - Select Install Type

Select Install Type

Please select install type and provide your AMD.com E-mail Address and password for authentication.

User Authentication

Please provide your AMD user account credentials to download the required files.
If you don't have an account, [please create one](#). If you forgot your password, you can [reset it here](#).

E-mail Address

Password

☒ Download and Install Now

Select your desired device and tool installation options and the installer will download and install just what is required.

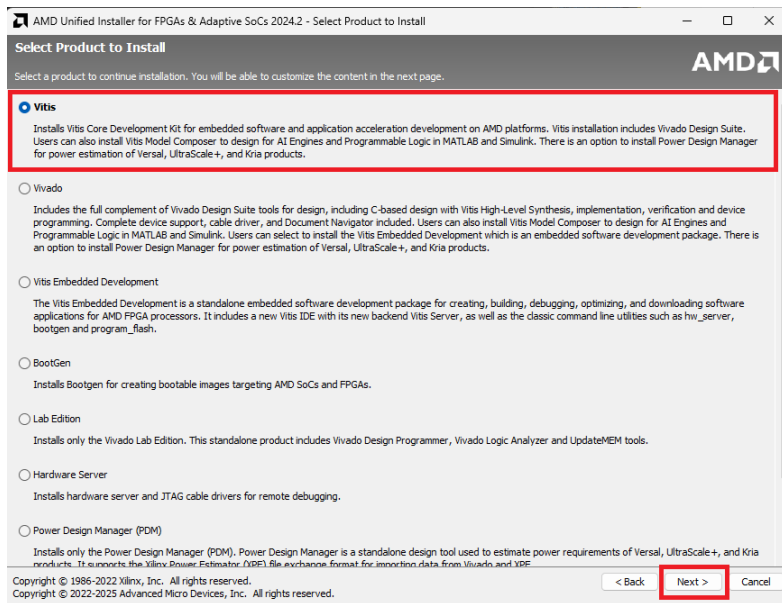
☐ Download Image (Install Separately)

The installer will download an image containing all devices and tool options for later installation. Use this option if you wish to install a full image on a network drive or allow different users maximum flexibility when installing.

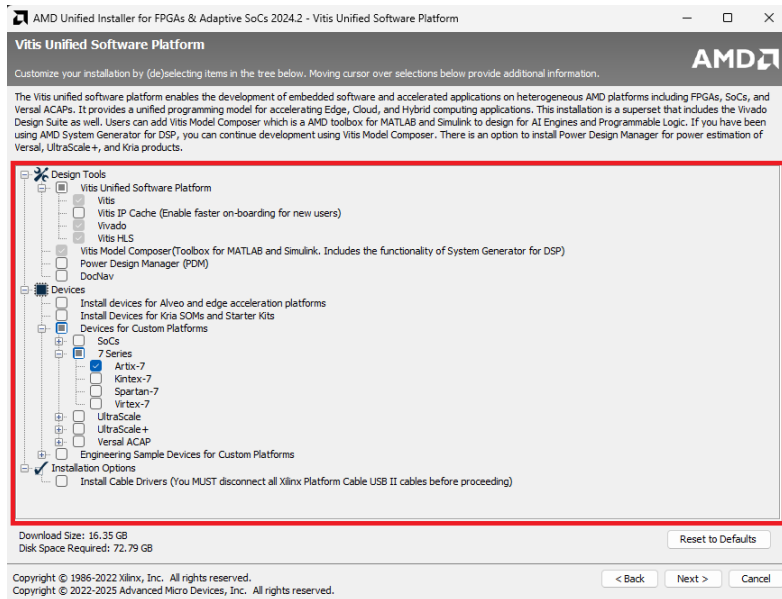
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< Back Next > Cancel

Instalación de Vitis/Vivado



Instalación de Vitis/Vivado



Instalación de Vitis/Vivado

AMD Unified Installer for FPGAs & Adaptive SoCs 2024.2 - Accept License Agreements

Accept License Agreements

Please read the following terms and conditions and indicate that you agree by checking the I Agree checkboxes.

End User License Agreement for Vitis

By checking "I Agree" below, or OTHERWISE ACCESSING, DOWNLOADING, INSTALLING or USING THE SOFTWARE, I AGREE on behalf of licensee to be bound by the agreement, which can be viewed by [clicking here](#).

☐ I Agree

End User License Agreement for Vivado

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☐ I Agree

End User License Agreement for Model Composer

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☐ I Agree

Third Party Software End User License Agreement for Vivado

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Instalación de Vitis/Vivado

AMD Unified Installer for FPGAs & Adaptive SoCs 2024.2 - Select Destination Directory

Select Destination Directory

Choose installation options such as location and shortcuts.

Installation Options

Select the installation directory

D:\Xilinx

Installation location(s)

D:\Xilinx\Vitis\2024.2

D:\Xilinx\Vivado\2024.2

D:\Xilinx\Vitis_HLS\2024.2

D:\Xilinx\Model_Composer\2024.2

Download location

D:\Xilinx\Downloads\Vitis_2024.2

Disk Space Required

Download Size:	16.35 GB
Disk Space Required:	72.79 GB
Final Disk Usage:	38.9 GB
Disk Space Available:	2024.09 GB

Select shortcut and file association options

☒ Create program group entries

Xilinx_Design_Tools

☒ Create desktop shortcuts

☒ Create file associations

Apply shortcut & file association selections to

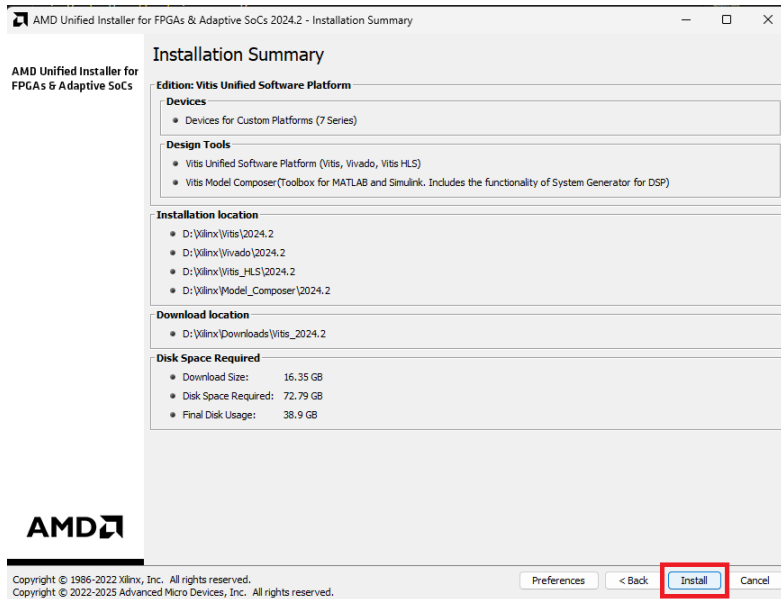
☒ Current user

☐ All users

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Instalación de Vitis/Vivado



Test de Vivado



Test de Vivado

- Incluir información del Kit de desarrollo en Vivado
 - Ubicar el directorio de instalación **Xilinx/Vivado/2024.2/data/boards/**
 - Descomprimir el archivo **board_files.zip**
 - Copiar el contenido de la carpeta **board_files** a **Xilinx/Vivado/2024.2/data/boards/board_files/**
 - Verificar que se encuentre la carpeta del kit a utilizar **Xilinx/Vivado/2024.2/data/boards/board_files/art7-a7-35**
- Generar binario de prueba
 - Ejecutar Vivado desde el acceso directo o terminal
 - Crear un proyecto nuevo
 - Seleccionar la placa Arty A7-35
 - Incluir los archivos top_microblaze.v, tb_top_microblaze.v y Arty-A7-35-Master.xdc
 - Simular el diseño y verificar el funcionamiento
 - Generar el binario
 - Los pasos mencionados se detallan en las figuras de las Pag. 16 - 33

NOTA

- En caso de tener problemas en la ejecución de la síntesis o simulación en Linux, es conveniente tener como lenguaje principal **en_US.UTF-8**.

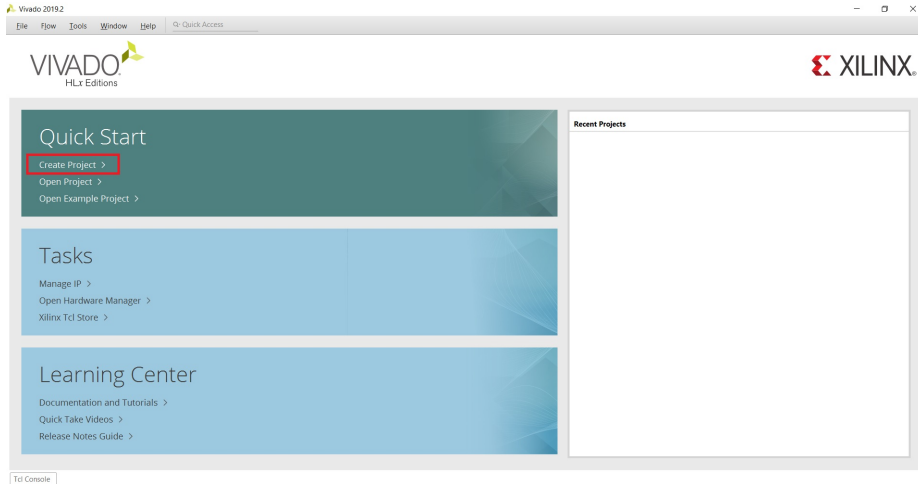
Opción 1 Habilitar **en_US.UTF-8** desde el archivo **locale**.

- sudo nano /etc/locale.conf

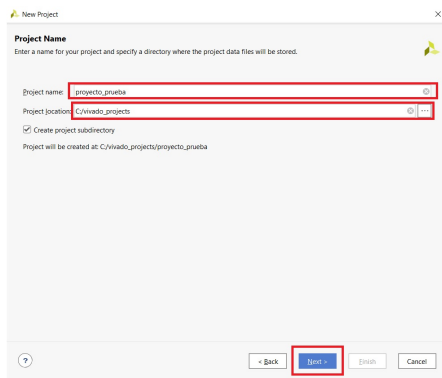
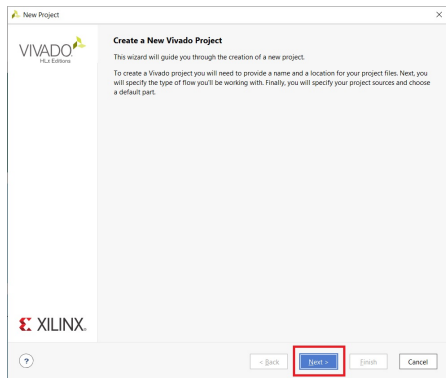
Opción 2 Ejecutar Vivado anteponiendo la definición del lenguaje.

- LC_ALL=en_US.UTF-8 /tools/Xilinx/Vivado/2024.2/bin/vivado

Test de Vivado



Test de Vivado



Test de Vivado

New Project

Project Type
Specify the type of project to create.

☒ **RTL Project**
You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
☒ Do not specify sources at this time

☐ **Post-synthesis Project**
You will be able to add sources, view device resources, run design analysis, planning and implementation.
☐ Do not specify sources at this time

☐ **I/O Planning Project**
Do not specify design sources. You will be able to view part/package resources.

☐ **Imported Project**
Create a Vivado project from a Synplify, XST or ISE Project File.

☐ **Example Project**
Create a new Vivado project from a predefined template.

[? < Back](#) [Next >](#) [Finish](#) [Cancel](#)

New Project

Default Part
Choose a default Xilinx part or board for your project.

Parts **Boards**

[Reset All Filters](#) [Update Board Repositories](#)

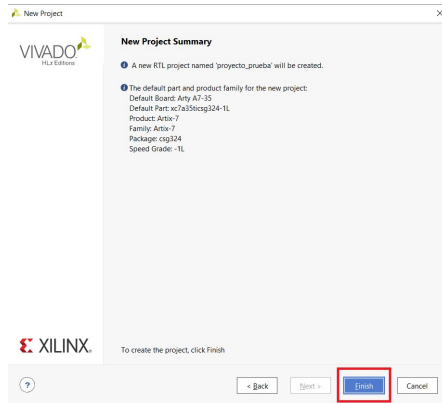
Vendor: All Name: All Board Rev: Latest

Search:

Display Name	Preview	Vendor	File Version	Part	I/O
Alpha-Data ADM-PCIE-7V3		alpha-data.com	1.1	xc7vx690tffg1157-2	115
Arty A7-100		digilentinc.com	1.0	xc7a100tcsg324-1	324
Arty A7-35		digilentinc.com	1.0	xc7a35tcsg324-1L	324
Arty S7-25		digilentinc.com	1.0	xc7s25csga324-1	324
Arty S7-50		digilentinc.com	1.0	xc7s50csga324-1	324

[? < Back](#) [Next >](#) [Finish](#) [Cancel](#)

Test de Vivado

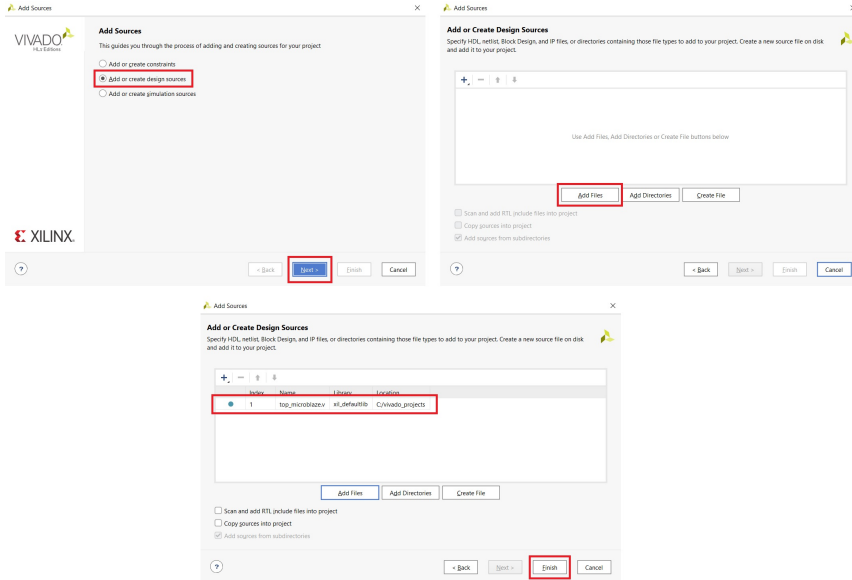


Test de Vivado

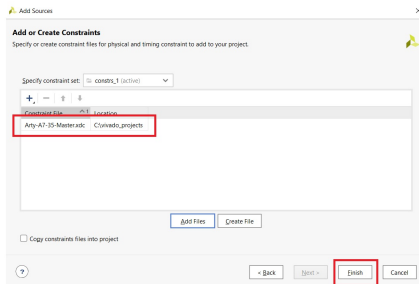
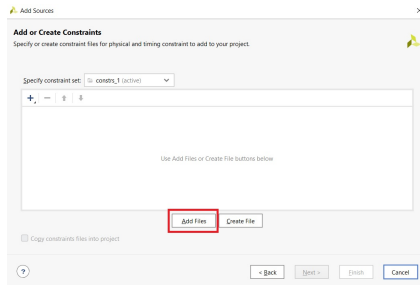
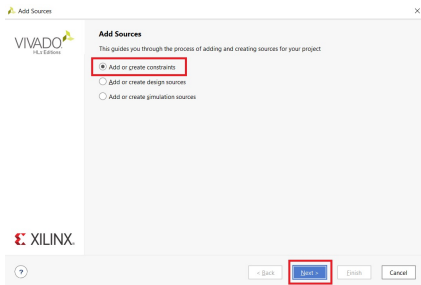
The screenshot displays the Vivado 2019.2 IDE interface for a project named 'proyecto_prueba'. The 'Flow Navigator' on the left shows the project structure, with 'Add Sources' highlighted under 'PROJECT MANAGER'. The 'PROJECT MANAGER' panel shows the 'Sources' tab with a tree view of Design Sources, Constraints, Simulation Sources, and Utility Sources. The 'Properties' panel is empty, showing 'Select an object to see properties'. The 'Project Summary' panel on the right shows the 'Overview' tab with project details: Project name: proyecto_prueba, Project location: C:/vivado_projects/proyecto_prueba, Product family: Artix-7, Project part: Arty A7-35 (xc7a35t1csg324-1L), Top module name: Not defined, Target language: Verilog, and Simulator language: Mixed. The 'Board Part' section shows Display name: Arty A7-35, Board part name: digilentinc.com:arty-a7-35:part0:1.0, and Board revision: E.0. The 'Design Runs' panel at the bottom shows a table of design runs.

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started															Vivado Synthesis Defaults (Vivado Synthesis 2019)
impl_1	constrs_1	Not started															Vivado Implementation Defaults (Vivado Implementation 2019)

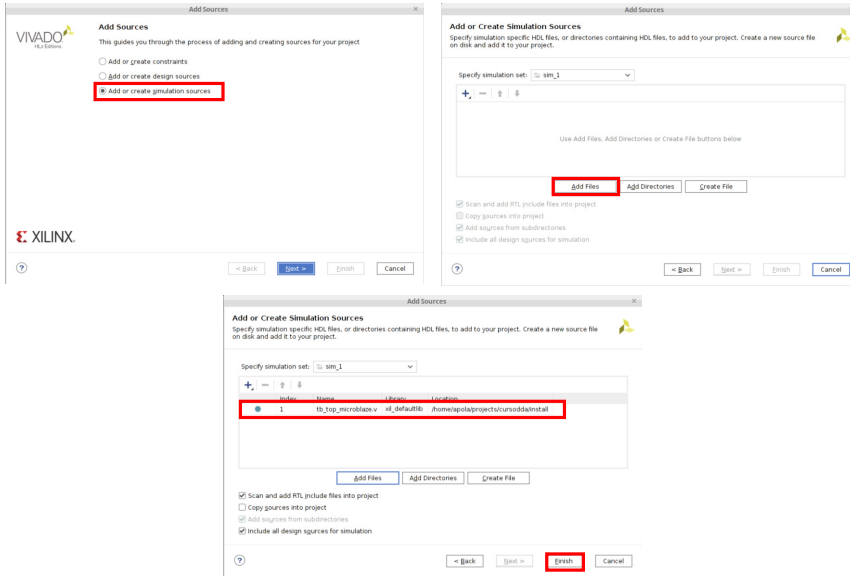
Test de Vivado



Test de Vivado



Test de Vivado



Test de Vivado

The screenshot displays the Vivado 2020.2 IDE interface. The top menu bar includes File, Edit, Flow, Tools, Repgrts, Window, Layout, View, and Help. The title bar shows the project path: install - [/home/apola/projects/run/install/install.xpr] - Vivado 2020.2. The status bar at the bottom indicates 'write_bitstream Complete' with a green checkmark.

The left sidebar contains the 'PROJECT MANAGER' section with the following options:

- Settings
- Add Sources
- Language Templates
- IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design

The main workspace is divided into three panels:

- Sources:** A red box highlights the 'Design Sources' section, which includes:
 - top_microblaze (top_microblaze.v)
 - Constraints (1)
 - constrs_1 (1)
 - Arty-A7-35-Master.xdc
 - Simulation Sources (1)
 - sim_1 (1)
 - tb_top_microblaze (tb_top_microblaze.v) (1)
 - u_top_microblaze : top_microblaze (top_microblaze.v)
 - Utility Sources

- Project Summary:** The 'Overview' tab is selected. It shows the following details:
- Project name: install
- Project location: /home/apola/projects/run/install
- Product family: Artix-7
- Project part: Arty A7-35 (xc7a35ticsg324-1L)
- Top module name: top_microblaze
- Target language: VHDL
- Simulator language: Mixed
- Design Runs:** The 'Tcl Console' tab is selected, showing a table of design runs.

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAM	URAM
synth_1	constrs_1	synth_design Complete!								9	5	0.0	0
impl_1	constrs_1	write_bitstream Complete!	7.387	0.000	0.243	0.000	0.000	0.065	0	8	5	0.0	0

Test de Vivado

The screenshot displays the Vivado 2020.2 IDE interface. The 'PROJECT MANAGER' tab is active, showing a project named 'install'. The 'Sources' pane lists the project's components, including 'top_microblaze' and 'tb_top_microblaze'. The 'Run Simulation' menu item is highlighted in the 'SIMULATION' section of the left-hand Project Navigator. A context menu is open, showing options for running behavioral, post-synthesis functional, post-synthesis timing, post-implementation functional, and post-implementation timing simulations. The 'Design Runs' pane at the bottom shows the status of the current run, indicating that the synthesis and bitstream writing are complete.

install - [/home/apola/projects/run/install/install.xpr] - Vivado 2020.2

write_bitstream Complete ✓

Default Layout

PROJECT MANAGER - install

PROJECT MANAGER

- Settings
- Add Sources
- Language Templates
- IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design
- SIMULATION
 - Run Simulation**
- RTL ANALYSIS
 - Open Elaborated Design
- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION
 - Run Implementation
 - Open Implemented Design

Properties

Sources

- Design Sources (1)
 - top_microblaze (top_microblaze.v)
- Constraints (1)
 - constrs_1 (1)
- Simulation Sources (1)
 - sim_1 (1)
 - tb_top_microblaze (tb_top_microblaze.v) (1)
 - u_top_microblaze : top_microblaze (top_microblaze.v)
- Utility Sources

Hierarchy Libraries Compile Order

Project Summary

Overview | Dashboard

Settings Edit

Project name: install

Project location: /home/apola/projects/run/install

Product family: Artix-7

Project part: Art7a35ticsg324-1L

Top module name: top_microblaze

Target language: VHDL

Simulator language: Mixed

Board Part

Log Reports Design Runs

Status WNS TNS WHS THS TPWS Total Power Failed Routes LUT FF BRAM URAM

synth_design Complete! 7.387 0.000 0.243 0.000 0.000 0.065 0 8 5 0.0 0

write_bitstream Complete!

Vivado Simulator

Test de Vivado

The screenshot displays the Vivado IDE interface during a behavioral simulation. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, Run, and Help. The 'Run' button is highlighted with a red box. The 'Flow Navigator' on the left shows the project structure, with 'SIMULATION' selected. The 'Scope' window shows the design hierarchy, and the 'Objects' window shows the current object. The 'Tcl Console' window shows the simulation progress and completion message.

Scope

Name	Design ...	Block T...
tb_tc	tb_top_mic	Verilog Mo
u_1	top_microt	Verilog Mo
gbl1	gbl1	Verilog Mo

Objects

Name	Value	Data Ty...
clock	0	Logic
i_sw[0]	0	Array
o_led	0	Array
uart_0	0	Logic
uart_0	0	Logic
reset	0	Logic

Untitled 1

Name	Value	999,999 ps	1,000,000 ps
clock	0		
i_sw[3:0]	0		
o_led[3:0]	0		
uart_rx_d_out	0		
uart_tx_d_in	0		
reset	0		

Tcl Console

```
xsim: Time (s): cpu = 00:00:17 : elapsed = 00:00:08 : Memory (MB): peak = 7427.797 : gain = 81.852 : free physical = 199 : free virtual = 4219
INFO: [USF-XSIm-96] XSIm completed. Design snapshot 'tb_top_microblaze_behav' loaded.
INFO: [USF-XSIm-97] XSIm simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:26 : elapsed = 00:00:18 : Memory (MB): peak = 7427.797 : gain = 92.816 : free physical = 199 : free virti
```

Sim Time: 1 us

Test de Vivado

The screenshot displays the Vivado IDE interface during a behavioral simulation. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, Run, and Help. The status bar at the top right indicates 'write_bitstream Complete' and 'Default Layout'.

The **Flow Navigator** on the left shows the project structure with sections for PROJECT MANAGER, IP INTEGRATOR, SIMULATION, RTL ANALYSIS, SYNTHESIS, and IMPLEMENTATION. The **SIMULATION** section is active, showing 'Run Simulation'.

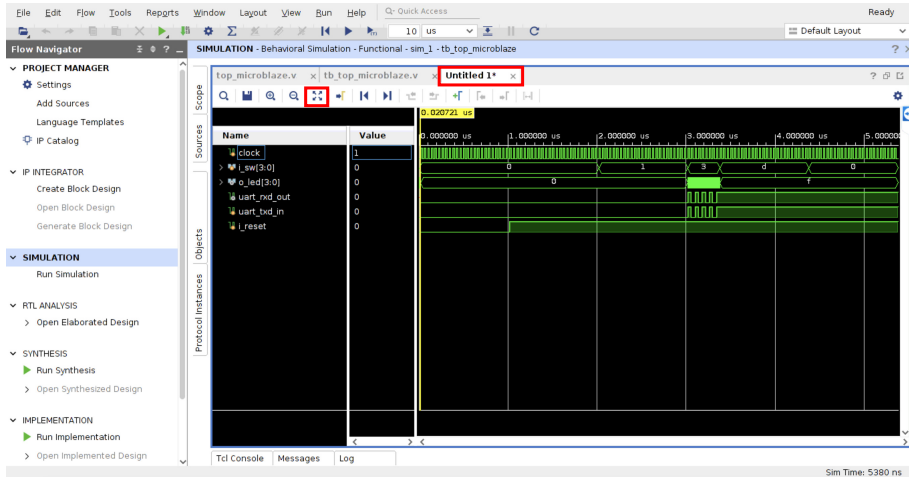
The main workspace is divided into three panes:

- Scope**: Lists design elements including 'tb_top_microblaze' and 'u_top_microblaze'.
- Objects**: A table showing the current scope's objects.

Name	Value	Data Type
clock	1	Logic
i_sw	0	Array
o_led	f	Array
uart_1		Logic
uart_1		Logic
l_reset	1	Logic
- Code Editor**: Displays the Verilog testbench code for 'tb_top_microblaze.v'. The code includes clock generation, signal initialization, and a call to 'top_microblaze'. A red box highlights the '\$finish;' statement at line 40.

The **Tcl Console** at the bottom shows the simulation results, including the command 'run all' and the message '\$finish called at time : 5380 ns'. The status bar at the bottom right indicates '40:1 Insert Sim Time: 5380 ns Verilog'.

Test de Vivado

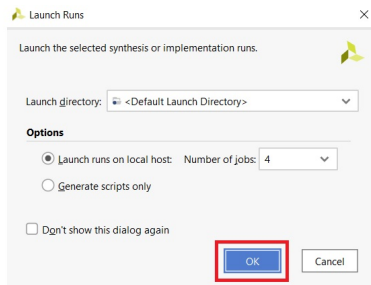
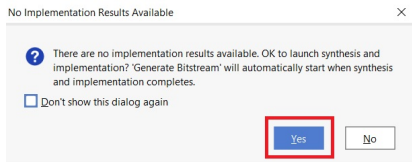


Test de Vivado

The screenshot displays the Vivado 2019.2 IDE interface for a project named 'proyecto_prueba'. The 'Flow Navigator' on the left shows the project workflow, with 'Generate Bitstream' highlighted under the 'PROGRAM AND DEBUG' section. The 'PROJECT MANAGER' pane shows the 'Sources' tab with a hierarchy of 'Design Sources' (top_microblaze), 'Constraints' (constrs_1), and 'Simulation Sources' (sim_1). The 'Properties' pane is empty, prompting the user to 'Select an object to see properties'. The 'Project Summary' pane on the right provides an overview of the project settings, including the project name, location, family (Artix-7), part (Artix-7-35), and target language (Verilog). The 'Design Runs' table at the bottom shows the status of the synthesis process.

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started															Vivado Synthesis Defaults (Vivado Synthesis 2019)
impl_1	constrs_1	Not started															Vivado Implementation Defaults (Vivado Implementation 2019)

Test de Vivado

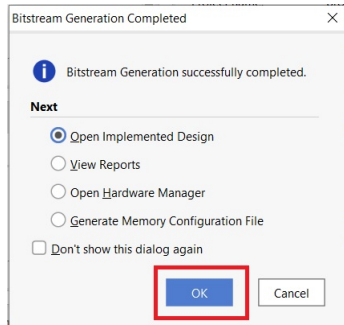


Test de Vivado

The screenshot displays the Vivado IDE interface for a project named 'proyecto_prueba'. The top menu bar includes File, Edit, Flow, Tools, Reports, Window, Layout, View, and Help. A red box highlights the 'Running synth_design' button in the top right corner, with a 'Cancel' button next to it. The left sidebar shows the 'Flow Navigator' with sections for IP INTEGRATOR, SIMULATION, RTL ANALYSIS, SYNTHESIS, and IMPLEMENTATION. The 'SYNTHESIS' section is expanded, showing 'Run Synthesis' and 'Open Synthesized Design'. The main workspace is divided into three panes: 'Sources' (showing Design Sources, Constraints, and Simulation Sources), 'Properties' (empty), and 'Project Summary' (showing Overview and Settings). The 'Design Runs' pane at the bottom shows a table of synthesis runs.

Name	Constraints	Status	WNS	TNS	WHS	THS	TPWS	Total Power	Failed Routes	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Running synth_design...													5/22/2	00:00:00	Vivado Synthesis Defaults (Vivado Synthesis 201
impl_1	constrs_1	Queued...														00:00:00	Vivado Implementation Defaults (Vivado Imple

Test de Vivado



Test de Vivado

The screenshot displays the Vivado IDE interface. On the left, the Flow Navigator shows the project hierarchy with 'IMPLEMENTATION' selected. The main workspace is divided into several panes: 'Sources' and 'Netlist' at the top left, 'Source File Properties' below them, and a large 'Timing' report pane at the bottom. The 'Timing' pane shows a 'Design Timing Summary' table with columns for Setup, Hold, and Pulse Width. The table indicates that all user-specified timing constraints are met. The background of the main workspace shows a circuit diagram with components labeled X0Y0, X0Y1, X0Y2, X1Y0, X1Y1, and X1Y2.

Flow Navigator

- SYNTHESIS
 - Run Synthesis
 - Open Synthesized Design
- IMPLEMENTATION**
 - Run Implementation
 - Open Implemented Design**
 - Constraints Wizard
 - Edit Timing Constraints
 - Report Timing Summary
 - Report Clock Networks
 - Report Clock Interaction
 - Report Methodology
 - Report DRC
 - Report Noise
 - Report Utilization
 - Report Power
 - Schematic
- PROGRAM AND DEBUG
 - Generate Bitstream
 - Open Hardware Manager

IMPLEMENTED DESIGN - xc7a35tscg324-1L

Sources | **Netlist**

- top_microblaze
 - Nets (35)
 - Leaf Cells (28)

Source File Properties

Arty-A7-35-Master.xdc

General | Properties

Timing

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 7.387 ns	Worst Hold Slack (WHS): 0.243 ns	Worst Pulse Width Slack (WPWS):
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS):
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints:
Total Number of Endpoints: 5	Total Number of Endpoints: 5	Total Number of Endpoints:

All user specified timing constraints are met.

Timing Summary - impl_1 (saved)